

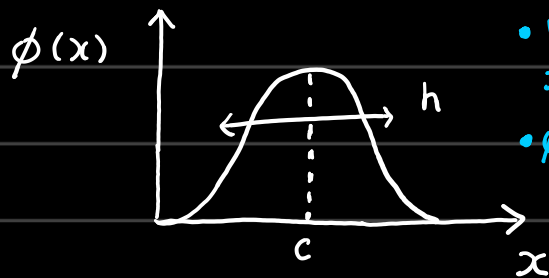
gaussian basis function:

$$\phi_j(x) = \exp \left\{ -\frac{(x - \mu_j)^2}{2s^2} \right\}$$

sigmoid basis function:

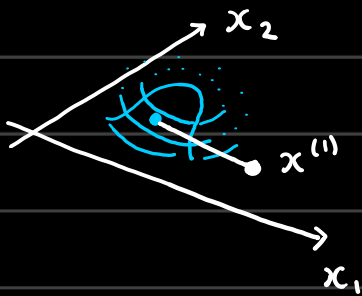
$$\phi_j(x) = \sigma \left(\frac{x - \mu_j}{s} \right)$$

in the 1-D case:



- We basically measure how far is x away from central parameter c .
- $\phi(x) \uparrow$ as $x \rightarrow c$

in the 2-D case:



We can have a family of RBFs:

